

# Madire Varun

Software Development Engineer — Backend & Distributed Systems  
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## Technical Skills

**Backend & Systems:** Distributed systems, microservices, concurrency, Kafka, caching, async processing, API design  
**Languages:** Java, C++, Python  
**Core CS:** Data structures & algorithms, scalability, high availability, consistency  
**Databases:** Oracle, MySQL, MongoDB, indexing, query optimization  
**Cloud & DevOps:** Azure, AWS, Jenkins, CI/CD, Git, Bitbucket, Control-M  
**Frameworks:** Spring Boot  
**Tools & Platforms:** Postman, Jira, Linux

## Experience

### Amadeus Labs

Bengaluru, India

#### Software Development Engineer

July 2023 – Present

- Designed and delivered scalable, fault-tolerant backend components for the Amadeus Hospitality Platform, processing **3K+** RPS and impacting **1M+** users.
- Optimized inventory flat file generation, reducing processing time by **50%**.
- Supported and monitored migration of the CRS application to Azure cloud.
- Optimized flat file generation of inventory, reducing processing time by **50%**.
- Improved shopping & pricing throughput by **30%** via partial blob optimization.
- Refactored legacy policy-override logic, improving maintainability and consistency.
- Modernized **20+** production services during on-call rotations, achieving **99.9%** uptime.
- Owned backend services & batch workflows ensuring SLA/SLO compliance; mentored junior engineers.
- Optimized SQL query performance using execution plan analysis and index tuning (removed unused indexes and redesigned inefficient indexes), reducing query latency by **25–30%** and lowering I/O overhead.
- Designed Kafka-based pipeline (**50K+** events/day) improving async throughput by **3×**.
- Implemented range-based partitioning on high-volume inventory tables, improving p95 query latency by **35–45%** and significantly reducing full table scans.
- Streamlined SQL by fixing N+1 patterns and rewriting hot-path queries, cutting **p95** by **15%** and reducing DB resources.

#### Software Development Engineer Intern

Feb 2023 – June 2023

- Built full-stack application integrating SMS, WhatsApp, and Email APIs for automated workflows.
- Developed Angular frontend and Spring Boot backend; delivered production-ready solution on time.

## Projects

### Personal AI Knowledge Engine | Python, FastAPI, React, FAISS/Pinecone, LLM, RAG

- Built a RAG-based AI assistant for querying personal documents and emails using natural language processing.
- Implemented vector embeddings and semantic search using FAISS/Pinecone for efficient information retrieval.
- Designed data ingestion pipelines with text chunking, metadata tagging, and indexing.
- Integrated LLMs (OpenAI) for context-aware response generation with reduced hallucinations.
- Applied metadata filtering and retrieval optimization for personalized results with local storage for data privacy.

## Education

### BV Raju Institute of Technology

Aug 2019 – June 2023

Computer Science and Engineering

Medak, India

## Achievements

**Game Changer Award-** Amadeus Labs: Recognized for exceptional ownership of a large-scale Cloud Migration initiative while simultaneously supporting feature teams, demonstrating cross-functional versatility and sustained commitment to delivery

**Going Above and Beyond Award-** Amadeus Labs: Honored for high-impact contributions to complex feature development and performance optimizations on time-critical platforms, consistently delivering under tight deadlines.